

Operating system

Project 2 Report

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1. 개발환경

Windows 11 Pro 22H2 / Oracle VM VirtualBox 7.0.10 / Ubuntu 18.04.2 / Linux kernel 4.20.11

2. CPU Scheduling과 Scheduling policies

현대의 운영체제는 대부분 multi-programming 방식을 채택하기 때문에, CPU Scheduling이 필연적으로 필요하다. CPU Scheduling이란 특정한 기준에 따라 CPU를 사용할 프로세스를 결정하는 것이다. 이러한 scheduling의 궁극적인 목적은 CPU의 유휴 시간을 줄이고 활용 시간을 늘려 성능을 올리는 데 있다. Scheduling은 크게 두 종류로 나눌 수 있다. Preemptive scheduling과 Non-preemptive scheduling이 그것인데, 전자의 경우 OS가 임의적으로 CPU를 사용하는 프로세스를 변경할 수 있는 반면, 후자는 프로세스가 스스로 CPU 사용을 완료할 때까지 간섭할 수 없다. Scheduling을 하는 세부적인 기법(정책)은 다양하다. 대표적으로는 First-Come, First-Served(FCFS), Shortest Job First(SJF), Round-Robin(RR), Shortest Remain Time First(SRTF) 등이 존재한다. 이들의 정책은 각각의 장단점이 존재하기에 정책을 평가하기 위한 별도의 기준이 필요한데, 대표적으로 CPU Utilization, Waiting time, Throughput, Turnaround time, Response time 등이 있다. 이러한 성능적 지표는 운영체제가 구동되는 시스템 및 환경에 따라 중요도가 달라지기 때문에 시스템을 정확히 파악하고 적합한 기준을 선택하여 그에 따른 Scheduling policy를 이용하여 Scheduler를 디자인 하는 것이 중요하다.

3. 구현

1. ku_CPU

네 가지 정책(FCFS, SJF, SRTF, RR) 각각을 기반으로 동작하는 syscall 4개를 구현하고 각각 335, 336, 337, 338번에 할당하였다.

(1) 기본 함수

ku_pop2() : Waiting Queue(이하 WQ)의 head에 있는 원소를 삭제하고 반환하는 함수.

ku_push2(job_t new) : 프로세스 정보를 parameter로 받아 queue에 삽입하는 함수.

ku_is_empty2() : WQ가 비어있는지 확인하는 함수.

ku_is_new2(pid_t chkpid) : 특정 pid를 가진 프로세스가 WQ에 존재하는지 확인하는 함수. WQ에 삽입하기 전 항상 이 함수를 이용해 중복 여부를 검사한다.

ku_compinsert2(job_t new) : 프로세스 정보를 삽입하고 job의 크기에 따라 queue를 정렬하는 함수. Head에 가장 작은 job값을 가진 프로세스가 오도록 정렬한다.

(2) FCFS

프로세스의 이름과 작업량(CPU burst)을 parameter로 받는다. 현재 CPU가 처리중인 프로세스가 없다면 (now == IDLE) 작업을 요청한 프로세스의 pid를 now에 할당한다.

(a) 현재 CPU에서 처리중인 프로세스가 CPU에 작업을 요청한 프로세스라면 요청한 프로세스의 job 값을 확인하고, 0이면 "Process Finish : (name)"을 출력한다. 이후 WQ가 비어있다면 IDLE 상태로 변경하고, 그렇지 않다면 다음 프로세스를 할당한다. 요청이 정상적으로 수락되었으므로, 최종 반환값은 1이다.

(b) 다른 프로세스가 작업을 요청했다면 해당 프로세스를 WQ에 push하고 "Working denied : (name)"을 출력한다. 요청이 거절되었으므로 반환값은 0이다.

(3) SJF

기본적인 골자는 FCFS와 동일하다. 그러나 현재 CPU가 처리하고 있는 프로세스가 CPU에 작업을 요청한 프로세스와 다를 경우, WQ에 프로세스를 삽입한 이후 job의 크기에 따라 오름차순으로 정렬하도록 변경했다(ku_compinsert2 이용).

(4) SRTF

SJF의 틀에서 now_job(int)이라는 변수를 추가하여 현재 CPU가 처리중인 프로세스의 남은 작업량을 관리할 수 있도록 하고, 이를 이용해 preemptive한 scheduling을 구현하였다.

작업을 요청한 프로세스가 CPU가 처리중인 프로세스일 경우, now_job의 값은 (프로세스가 전달한 값 - 1)이 된다 (0일 경우 제외). CPU가 처리하고 있지 않은 프로세스가 작업을 요청했을 경우에는 현재 CPU가 처리하고 있는 프로세스의 남은 작업량 (now_job)과 작업을 요청한 프로세스의 작업량을 비교하여, 후자가 더 작을 경우 CPU가 처리하는 프로세스를 교체한다. 그렇지 않을 경우엔 WQ에 해당 프로세스를 삽입하고 job의 크기에 대해 오름차순으로 정렬한다.

(5) RR

FCFS의 틀에서 timeslice(int)라는 변수를 도입하여 처리중인 프로세스의 잔여 time slice를 관리할 수 있도록 변경하였다. RR은 time slice의 길이 설정 또한 유의미한 차이를 가져오기에, timeslice 값은 10(1초)과 5(0.5초)로 설정하여 두 번 실험하였다.

CPU가 처리중인 프로세스가 작업을 요청했을 경우, 두가지를 확인하게 된다. 첫번째는 잔여 작업량으로, 만약 0이라면 timeslice를 원래의 값으로 되돌리고 프로세스를 교체한다. 두번째는 timeslice의 값으로, timeslice가 0이라면 "---> Turn over : (name)"

을 출력하고 프로세스를 교체한다. 두 경우 모두 해당되지 않는다면 timeslice를 1 감소시키고 최종적으로 0을 반환한다. 만약 CPU가 현재 처리하는 프로세스가 아닌 프로세스가 작업을 요청한다면, 별다른 처리 없이 WQ에 삽입하고 최종적으로 1을 반환한다.

2. User application (Process)

실행시 job, delay, name을 받아 0.1초마다 ku_cpu syscall을 호출하여 반환 값이 1일 경우 (작업 요청 수락) job을 감소시키고, 0일 경우 wait을 증가시킨다. 이후 작업이 완료(job = 0)되면 마지막으로 CPU가 해당 프로세스에 대한 작업 완수 동작을 수행하도록 syscall을 시행하고, 총 wait time을 출력한다.

4개의 정책을 테스트하기 위해 동일한 코드에 syscall 번호만 다르게 하여 4개의 파일(p.c, p2.c, p3.c, p4.c)을 생성하였다. 이후 다음과 같은 표의 시나리오를 따라 프로세스 4개를 실행하고 각각의 wait time을 산출해 분석했다. 이후 추가적으로 response time에 대한 분석도 시행했다.

Process	Arrival time (Delay)	Burst time
A	0	7
B	2	4
C	4	1
D	5	4

Table 1. Scenario of process execution

4. 결과

1. 실험 결과

1-1. FCFS

- User Application

```
kmr@kmr-VirtualBox:~$ sudo ./run
Process A : I will use CPU by 7s.
Process B : I will use CPU by 4s.
Process C : I will use CPU by 1s.
Process D : I will use CPU by 4s.
Process A : Finish! My total wait time is 0s.
Process B : Finish! My total wait time is 5s.
Process C : Finish! My total wait time is 7s.
Process D : Finish! My total wait time is 7s.
```

- Kernel Log(dmesg)

[49972.084937]	Working: A	[49974.493020]	Working Denied: B
[49972.185100]	Working: A	[49974.510682]	Working: A
[49972.286185]	Working: A	[49974.593712]	Working Denied: B
[49972.386705]	Working: A	[49974.610883]	Working: A
[49972.486973]	Working: A	[49974.694447]	Working Denied: B
[49972.587493]	Working: A	[49974.711128]	Working: A
[49972.688110]	Working: A	[49974.794537]	Working Denied: B
[49972.789427]	Working: A	[49974.812292]	Working: A
[49972.859012]	Working Denied: B	[49974.858543]	Working Denied: C
[49972.889859]	Working: A	[49974.894573]	Working Denied: B
[49972.959586]	Working Denied: B	[49974.913254]	Working: A
[49972.990468]	Working: A	[49974.959347]	Working Denied: C
[49973.060138]	Working Denied: B	[49974.995258]	Working Denied: B
[49973.090841]	Working: A	[49975.014221]	Working: A
[49973.165468]	Working Denied: B	[49975.060464]	Working Denied: C
[49973.191088]	Working: A	[49975.095625]	Working Denied: B
[49973.266160]	Working Denied: B	[49975.115738]	Working: A
[49973.291898]	Working: A	[49975.163388]	Working Denied: C
[49973.366405]	Working Denied: B	[49975.196744]	Working Denied: B

1-2. SJF

- User Application

```
kmr@kmr-VirtualBox:~$ sudo ./run2
Process A : I will use CPU by 7s.
Process B : I will use CPU by 4s.
Process C : I will use CPU by 1s.
Process D : I will use CPU by 4s.
Process A : Finish! My total wait time is 0s.
Process C : Finish! My total wait time is 3s.
Process B : Finish! My total wait time is 6s.
Process D : Finish! My total wait time is 7s.
```

- Kernel Log

[50131.877927]	Working Denied: B	[50132.934550]	Working: C
[50131.913703]	Working: A	[50132.953902]	Working Denied: D
[50131.923937]	Working Denied: C	[50132.994729]	Working Denied: B
[50131.945581]	Working Denied: D	[50133.035016]	Working: C
[50131.978544]	Working Denied: B	[50133.055422]	Working Denied: D
[50132.015684]	Working: A	[50133.097526]	Working Denied: B
[50132.025096]	Working Denied: C	[50133.135309]	Process Finish: C
[50132.045698]	Working Denied: D	[50133.155930]	Working Denied: D
[50132.078619]	Working Denied: B	[50133.199650]	Working: B
[50132.118552]	Process Finish: A	[50133.257102]	Working Denied: D
[50132.125621]	Working: C	[50133.300143]	Working: B
[50132.146292]	Working Denied: D	[50133.358685]	Working Denied: D
[50132.184700]	Working Denied: B	[50133.404683]	Working: B
[50132.226919]	Working: C	[50133.459705]	Working Denied: D
[50132.247547]	Working Denied: D	[50133.505755]	Working: B
[50132.285743]	Working Denied: B	[50133.563770]	Working Denied: D
[50132.327038]	Working: C	[50133.607039]	Working: B

1-3. SRTF

- User Application

```

kmr@kmr-VirtualBox:~$ sudo ./run3
Process A : I will use CPU by 7s.
Process B : I will use CPU by 4s.
Process C : I will use CPU by 1s.
Process D : I will use CPU by 4s.
Process C : Finish! My total wait time is 0s.
Process B : Finish! My total wait time is 1s.
Process D : Finish! My total wait time is 2s.
kmr@kmr-VirtualBox:~$
Process A : Finish! My total wait time is 9s.

```

- Kernel Log

[50231.404683] Working: A	[50234.008476] Working: B	[50236.041972] Working: B
[50231.507223] Working: A	[50234.112417] Working: B	[50236.743063] Working: B
[50231.609003] Working: A	[50234.187277] Working: C	[50236.843891] Working: B
[50231.710103] Working: A	[50234.214036] Working: Denied: B	[50236.944069] Working: B
[50231.811150] Working: A	[50234.288100] Working: C	[50237.045162] Working: B
[50231.911489] Working: A	[50234.389258] Working: C	[50237.149248] Working: B
[50232.013640] Working: A	[50234.492771] Working: C	[50237.250349] Process Finish: B
[50232.114295] Working: A	[50234.595971] Working: C	[50237.306429] Working: D
[50232.188003] Working: B	[50234.699181] Working: C	[50237.406549] Working: D
[50232.214960] Working Denied: A	[50234.801456] Working: C	[50237.507304] Working: D
[50232.289190] Working: B	[50234.903046] Working: C	[50237.608013] Working: D
[50232.391291] Working: B	[50235.008326] Working: C	[50237.708574] Working: D
[50232.491431] Working: B	[50235.109093] Working: C	[50237.808948] Working: D
[50232.592596] Working: B	[50235.187314] Working Denied: D	[50237.909143] Working: D
[50232.692985] Working: B	[50235.209626] Process Finish: C	[50238.011267] Working: D
[50232.795818] Working: B	[50235.222388] Working: B	[50238.111397] Working: D
[50232.897243] Working: B	[50235.327048] Working: B	[50238.212507] Working: D

1-4. RR

- User Application

kmr@kmr-VirtualBox:~\$ sudo ./run4	kmr@kmr-VirtualBox:~\$ sudo ./run4
Process A : I will use CPU by 7s.	Process A : I will use CPU by 7s.
Process B : I will use CPU by 4s.	Process B : I will use CPU by 4s.
Process C : I will use CPU by 1s.	Process C : I will use CPU by 1s.
Process D : I will use CPU by 4s.	Process D : I will use CPU by 4s.
Process C : Finish! My total wait time is 1s.	Process C : Finish! My total wait time is 2s.
Process B : Finish! My total wait time is 7s.	Process B : Finish! My total wait time is 8s.
Process A : Finish! My total wait time is 9s.	Process A : Finish! My total wait time is 10s.
Process D : Finish! My total wait time is 8s.	Process D : Finish! My total wait time is 9s.

(좌 : timeslice = 1 sec / 우 : timeslice = 0.5 sec)

- Kernel Log(timeslice = 1 sec)

```

[ 791.382488] Working: A [ 802.287521] Working: D
[ 791.483344] Working: A [ 802.389186] Working: D
[ 791.583946] Working: A [ 802.490502] ----> Turn Over: D
[ 791.687413] Working: A [ 802.536344] Working: B
[ 791.787579] ----> Turn Over: A [ 802.636518] Working: B
[ 791.894309] Working: A [ 802.741355] Working: B
[ 791.995766] Working: A [ 802.845360] Working: B
[ 792.106099] Working: A [ 802.949259] Working: B
[ 792.211115] Working: A [ 803.049709] Working: B
[ 792.312093] Working: A [ 803.150658] Working: B
[ 792.412318] Working: A [ 803.257789] Working: B
[ 792.514227] Working: A [ 803.360179] Working: B
[ 792.618648] Working: A [ 803.460407] Working: B
[ 792.724294] Working: A [ 803.561658] Process Finish: B
[ 792.824845] Working: A [ 803.567012] Working: A
[ 792.925542] ----> Turn Over: A [ 803.681568] Working: A
[ 792.982509] Working: B [ 803.784431] Working: A
[ 793.083242] Working: B [ 803.885918] Working: A
[ 793.184280] Working: B [ 803.989343] Working: A
[ 793.286480] Working: B [ 804.089938] Working: A
[ 793.387104] Working: B [ 804.192592] Working: A
[ 793.487394] Working: B [ 804.294548] Working: A
[ 793.587669] Working: B [ 804.399617] Working: A
[ 793.687877] Working: B [ 804.500103] Working: A
[ 793.788588] Working: B [ 804.601800] ----> Turn Over: A
[ 793.889040] Working: B [ 804.628375] Working: D
[ 793.989201] ----> Turn Over: B [ 804.729551] Working: D
[ 794.036900] Working: A [ 804.830318] Working: D
[ 794.137082] Working: A [ 804.937809] Working: D
[ 794.248574] Working: A [ 805.038134] Working: D
[ 794.349916] Working: A [ 805.139089] Working: D
[ 794.453174] Working: A [ 805.246581] Working: D
[ 794.557633] Working: A [ 805.347548] Working: D
[ 794.662100] Working: A [ 805.447767] Working: D
[ 794.766567] Working: A [ 805.548706] Working: D
[ 794.871034] Working: A [ 805.649733] ----> Turn Over: D
[ 794.975501] Working: A [ 805.730397] Working: A
[ 795.079968] Working: A [ 805.831060] Working: A
[ 795.184435] Working: A [ 805.931722] Working: A
[ 795.288902] Working: A [ 806.032383] Working: A
[ 795.393369] Working: A [ 806.133044] Working: A
[ 795.497836] Working: A [ 806.233705] Working: A
[ 795.602303] Working: A [ 806.334366] Working: A
[ 795.706770] Working: A [ 806.435027] Working: A
[ 795.811237] Working: A [ 806.535688] Working: A
[ 795.915704] Working: A [ 806.636349] Working: A
[ 796.020171] Working: A [ 806.737010] Working: A
[ 796.124638] Working: A [ 806.837671] Working: A
[ 796.229105] Working: A [ 806.938332] Working: A
[ 796.333572] Working: A [ 807.038993] Working: A
[ 796.438039] Working: A [ 807.139654] Working: A
[ 796.542506] Working: A [ 807.240315] Working: A
[ 796.646973] Working: A [ 807.340976] Working: A
[ 796.751440] Working: A [ 807.441637] Working: A
[ 796.855907] Working: A [ 807.542298] Working: A
[ 796.960374] Working: A [ 807.642959] Working: A
[ 797.064841] Working: A [ 807.743620] Working: A
[ 797.169308] Working: A [ 807.844281] Working: A
[ 797.273775] Working: A [ 807.944942] Working: A
[ 797.378242] Working: A [ 808.045603] Working: A
[ 797.482709] Working: A [ 808.146264] Working: A
[ 797.587176] Working: A [ 808.246925] Working: A
[ 797.691643] Working: A [ 808.347586] Working: A
[ 797.796110] Working: A [ 808.448247] Working: A
[ 797.900577] Working: A [ 808.548908] Working: A
[ 798.005044] Working: A [ 808.649569] Working: A
[ 798.109511] Working: A [ 808.750230] Working: A
[ 798.213978] Working: A [ 808.850891] Working: A
[ 798.318445] Working: A [ 808.951552] Working: A
[ 798.422912] Working: A [ 809.052213] Working: A
[ 798.527379] Working: A [ 809.152874] Working: A
[ 798.631846] Working: A [ 809.253535] Working: A
[ 798.736313] Working: A [ 809.354196] Working: A
[ 798.840780] Working: A [ 809.454857] Working: A
[ 798.945247] Working: A [ 809.555518] Working: A
[ 799.049714] Working: A [ 809.656179] Working: A
[ 799.154181] Working: A [ 809.756840] Working: A
[ 799.258648] Working: A [ 809.857501] Working: A
[ 799.363115] Working: A [ 809.958162] Working: A
[ 799.467582] Working: A [ 810.058823] Working: A
[ 799.572049] Working: A [ 810.159484] Working: A
[ 799.676516] Working: A [ 810.260145] Working: A
[ 799.780983] Working: A [ 810.360806] Working: A
[ 799.885450] Working: A [ 810.461467] Working: A
[ 799.989917] Working: A [ 810.562128] Working: A
[ 800.094384] Working: A [ 810.662789] Working: A
[ 800.198851] Working: A [ 810.763450] Working: A
[ 800.303318] Working: A [ 810.864111] Working: A
[ 800.407785] Working: A [ 810.964772] Working: A
[ 800.512252] Working: A [ 811.065433] Working: A
[ 800.616719] Working: A [ 811.166094] Working: A
[ 800.721186] Working: A [ 811.266755] Working: A
[ 800.825653] Working: A [ 811.367416] Working: A
[ 800.930120] Working: A [ 811.468077] Working: A
[ 801.034587] Working: A [ 811.568738] Working: A
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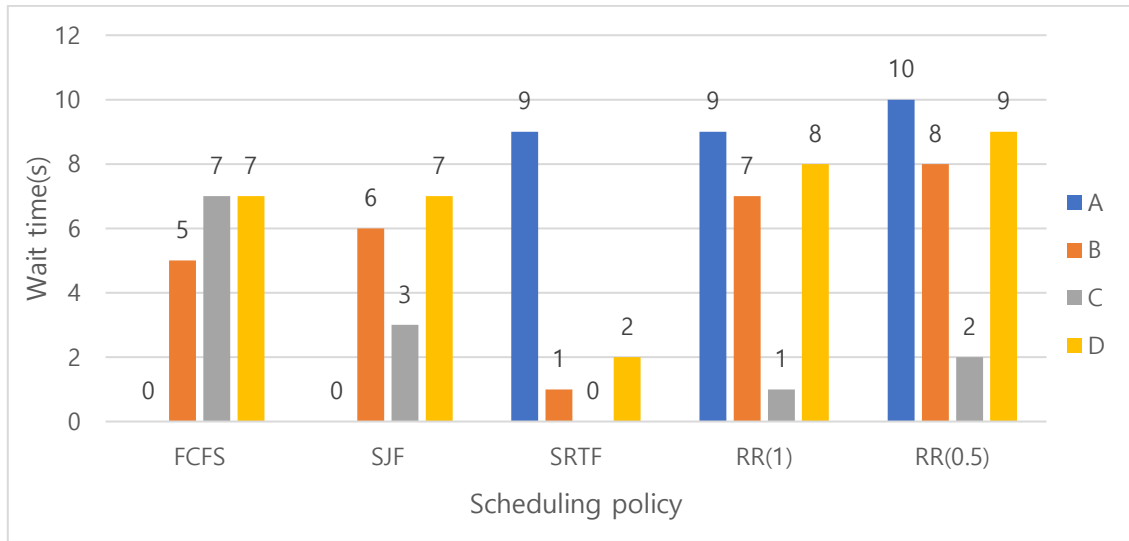


Fig 2. Wait times of four processes under four scheduling policies.

네 프로세스 각각의 wait time을 정책별로 비교했을 때 특징적으로 확인할 수 있는 점은 다음과 같다.

1. CPU burst가 긴 프로세스(A)의 경우 비선점형(FCFS, SJF)에 비해 선점형(SRTF, RR)에서 wait time이 높게 나타나고, CPU burst가 짧은 프로세스(C)는 그와 반대의 양상을 보였다.
2. SRTF의 경우 길이가 가장 길었던 A가 다른 세 프로세스에 비해 비교적 wait time이 매우 높았다.
3. RR의 경우 time slice가 짧아짐에 따라 모든 프로세스의 wait time이 전체적으로 증가했다.

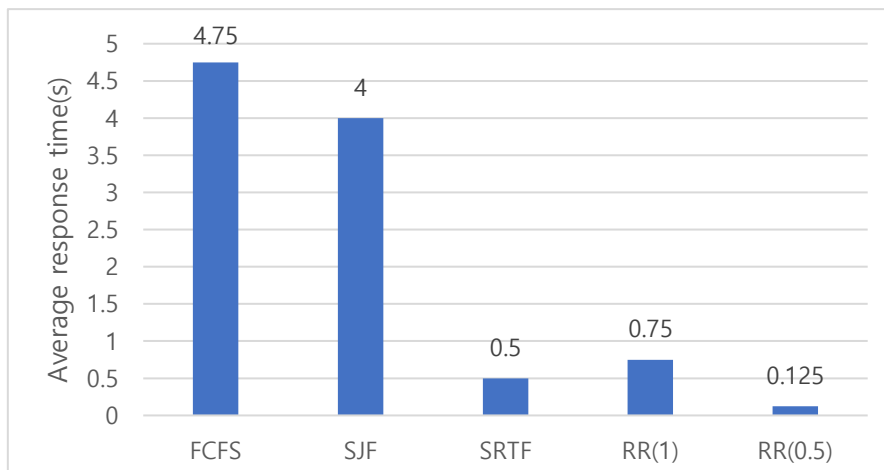


Fig 3. Average response time of four scheduling policies.

추가적으로 평균 response time을 비교했을 때, 가장 낮은 것은 time slice가 0.5sec인 RR이었고, 그 다음은 SRTF였다. 그 뒤로는 time slice가 1sec인 RR, SJF, FCFS순으로 평균 response time이 짧았다. 통상적으로 알려진 것과 달리 SRTF의 response time이 눈에 띄게 낮은데, 이는 실험에서 사용한 시나리오가 대부분의 작업 시간이 긴 프로세스의 요청이 짧은 프로세스보다 먼저 CPU에 도

착하기에 도출된 결과로 보인다.

종합하여 각 정책을 평가해보면 다음과 같다.

	장점	단점
FCFS	Context switching으로 인한 overhead가 낮음	Convoy effect 발생 가능
SJF	Wait time이 상대적으로 낮음	Starvation 발생 가능, CPU Burst의 정확한 계산의 어려움
SRTF	Wait time이 낮음	SJF와 단점 공유, context switching으로 인한 overhead가 상대적으로 높음
RR	Response time이 낮음(time slice 길이에 반비례)	잘은 context switching으로 overhead가 높음

Table 2. Pros & cons of four scheduling policies

위의 장단점을 고려하였을 때, 해당 정책이 사용되기 적합한 상황을 정리해보면 다음과 같다.

1. FCFS

Context switching으로 인한 overhead가 낮기 때문에, 모든 프로세스의 작업을 완료하는 것이 중요시되는 시스템에 사용될 수 있을 것이다. 또한, 대부분의 프로세스의 길이가 짧고, real-time의 중요성이 떨어지는 환경에서 사용될 수 있다.

2. SJF / SRTF

CPU Burst time을 정확히 측정하는 것이 관건인 정책이므로, 실행되는 프로세스가 크게 변하지 않고 각각의 Burst time이 뚜렷한 환경에서 사용될 수 있다.

3. RR

응답성이 좋은 정책이므로, real-time 기반의 시스템에서 사용될 수 있을 것이다. 다만 이를 위해선 시스템에서 사용되기 가장 적합한 time slice 크기를 결정할 필요가 있다.

5. 어려웠던 점 & 해결

- 초기 커널 컴파일 과정에서 다양한 오류가 발생했는데, 그 중 대부분이 아래와 같았다.

```
kernel/ku_cpu.c: In function 'ku_is_new2':
kernel/ku_cpu.c:67:2: error: 'for' loop initial declarations are only allowed in C99 or C11 mode
  for (int i = (wq.first + 1); i <= wq.last; i = (i + 1) % MAX) {
  ^~~~~
```

관련 정보를 찾아보니 리눅스 커널이 ISO C89표준을 따르도록 작성되었기 때문임을 확인할 수 있었다. C89 표준에 대해서는 지식이 전무했기 때문에 오류 메시지를 일일이 검색해가며 수정해 문제를 해결했다.

- SJF, SRTF를 구현할 때 waiting queue에 프로세스 삽입 후 작업량 순으로 정렬하는 것을 큐를 순회하며 크기를 비교하여 삽입하는 일종의 삽입 정렬의 형태로 구현하려고 하였으나, 해당 작업을 수행할 때 ku_cpu 내에서 무한 반복이 일어나는 문제가 발생하였다. Waiting queue를 원형 큐의 구조로 작성하였기에 이와 관련하여 문제가 발생했을 것으로 추정되나, 정확한 원인을 파악하는데 어려움이 있어 결국 성능을 일부 포기하고 큐에 데이터를 삽입한 이후 bubble sort하는 방식으로 해결하였다.